

AERO40006 - Mathematics 1

Tutorial 2

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Question 1

Evaluate these integrals using a change of variables (including the option of trigonometric substitutions),

(1)	$\int \sin(7x+1)dx$	(2)	$\int \cot(x)dx$
(3)	$\int \frac{3x^2-6x}{x^3-3x^2+1}dx$	(4)	$\int \frac{1}{\sqrt{1-x^2}}dx$

(1) Let $u = 7x+1$, $\frac{du}{dx} = 7$ $\therefore dx = \frac{1}{7}du$

$$\therefore \int \sin(7x+1) dx = \int \frac{\sin u}{7} du = -\frac{1}{7} \cos u + C = -\frac{1}{7} \cos(7x+1) + C$$

(2) $\int \cot(x) dx = \int \frac{1}{\tan(x)} dx = \int \frac{\cos(x)}{\sin(x)} dx$

Let $u = \sin(x)$ $\therefore \frac{du}{dx} = \cos(x)$ $\therefore dx = \frac{du}{\cos(x)}$

$$\therefore \int \cot(x) dx = \int \frac{1}{\sin(x)} du = \int \frac{1}{u} du = \ln|u| + C = \ln|\sin(x)| + C$$

(3) Let $u = x^3 - 3x^2 + 1$ $\therefore \frac{du}{dx} = 3x^2 - 6x$ $\therefore dx = \frac{du}{3x^2 - 6x}$

$$\therefore \int \frac{3x^2-6x}{x^3-3x^2+1} dx = \int \frac{1}{u} du = \ln|u| + C = \ln|x^3-3x^2+1| + C$$

(4) Let $x = \sin \theta$ $\therefore \frac{dx}{d\theta} = \cos \theta$ $\therefore dx = \cos \theta d\theta$

$$\therefore \int \frac{\cos \theta}{\sqrt{1-\sin^2 \theta}} d\theta = \int 1 d\theta = \theta + C = \arcsin x + C$$

Question 2

Evaluate these integrals using integration by parts,

(1)	$\int 2xe^x dx$	(2)	$\int x \sin(x) dx$
(3)	$\int x^2 \ln(x) dx$	(4)	$\int x \cos(3x) dx$

$$\begin{array}{lll}
 (1) & D & I \\
 & + & 2x \quad e^x \\
 & - & 2 \quad e^x \\
 & + & 0 \quad e^x
 \end{array}$$

$$\therefore \int 2xe^x dx = 2xe^x - 2e^x + C$$

$$\begin{array}{lll}
 (2) & D & I \\
 & + & x \quad \sin(x) \\
 & - & 1 \quad -\cos(x) \\
 & + & 0 \quad -\sin(x)
 \end{array}$$

$$\therefore \int x \sin(x) dx = -x \cos(x) + \sin(x) + C$$

$$\begin{array}{lll}
 (3) & D & I \\
 & + & \ln(x) \quad x^2 \\
 & - & \frac{1}{x} \quad \frac{1}{3}x^3 \\
 & + & -\frac{1}{x^2} \quad \frac{1}{12}x^4 \\
 & - & \dots
 \end{array}$$

$$\begin{aligned}
 \therefore \int x^2 \ln(x) dx &= \frac{1}{3}x^3 \ln(x) - \int \frac{1}{3}x^2 dx \\
 &= \frac{1}{3}x^3 \ln(x) - \frac{1}{12}x^3 + C
 \end{aligned}$$

$$\begin{array}{lll}
 (4) & D & I \\
 & + & x \quad \cos(3x) \\
 & - & 1 \quad \frac{1}{3} \sin(3x) \\
 & + & 0 \quad -\frac{1}{9} \cos(3x)
 \end{array}$$

$$\therefore \int x \cos(3x) dx = \frac{1}{3}x \sin(3x) + \frac{1}{9} \cos(3x) + C$$

Question 3

Use partial fractions, evaluate the following integrals,

(1) $\int \frac{3x+4}{(x+1)(x+2)} dx$	(2) $\int \frac{x+1}{(x-1)^2} dx$	(3) $\int \frac{5x^2}{(x^2+1)(2x-1)} dx$
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$$(1) \int \frac{3x+4}{(x+1)(x+2)} dx = \int \frac{1}{x+1} dx + \int \frac{2}{x+2} dx$$

$$= \ln|x+1| + 2\ln|x+2| + C$$

$$(2) \int \frac{x+1}{(x-1)^2} dx = \int \frac{2}{(x-1)^2} dx + \int \frac{1}{x-1} dx$$

$$= \frac{-2}{(x-1)} + \ln|x-1| + C$$

$$(3) \int \frac{5x^2}{(x^2+1)(2x-1)} dx = \int \frac{5}{2x-1} dx + \int \frac{-5}{2x^3-x^2+2x-1} dx$$

$$= \frac{5}{2} \ln|2x-1| - \frac{5}{6x^2-2x+2} \ln|2x^3-x^2+2x-1| + C$$

Question 4

Evaluate the following definite integrals:

(1)	$\int_0^1 (3x + 4)^{-3} dx$	(2)	$\int_1^2 x (1 + x^2)^{1/2} dx$
(3)	$\int_0^\pi \cos(4x) dx$	(4)	$\int_0^{\pi/2} \sin^2(x) \cos(x) dx$

$$(1) \int_0^1 (3x+4)^{-3} dx = \left[-\frac{1}{6} (3x+4)^{-2} \right]_0^1 = \left(-\frac{1}{244} \right) - \left(-\frac{1}{96} \right) = \frac{11}{1568}$$

$$(2) \quad u = 1+x^2 \quad \therefore \frac{du}{dx} = 2x \quad \therefore dx = \frac{du}{2x}$$

$$\int_1^2 x (1+x^2)^{\frac{1}{2}} dx = \int_1^5 \frac{1}{2} u^{\frac{1}{2}} du = \frac{1}{2} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_1^5 = 2.78$$

$$(3) \int_0^\pi \cos(4x) dx = \left[\frac{1}{4} \sin(4x) \right]_0^\pi = 0 - 0 = 0$$

$$(4) \quad u = \sin(x) \quad \therefore \frac{du}{dx} = \cos(x) \quad \therefore dx = \frac{du}{\cos(x)}$$

$$\int_0^{\frac{\pi}{2}} \sin^2(x) \cos(x) dx = \int_0^1 u^2 du = \left[\frac{1}{3} u^3 \right]_0^1 = \frac{1}{3}$$

Question 5

Evaluate the following integrals that involve recursive relations.

1. Let

$$I_m = \int_0^\pi \cos^m(x) dx, \quad n = 0, 1, 2, \dots$$

Derive a formula relating I_m and I_{m-2} (for $m \geq 2$). Use this formula to calculate I_4 .

2. Given that

$$I_n = \int_0^\pi e^x \sin^n(x) dx, \quad n = 0, 1, 2, \dots$$

show that

$$(n^2 + 1)I_n = n(n-1)I_{n-2}.$$

Verify that

$$I_5 = \frac{3}{13}(e^\pi + 1).$$

$$\begin{aligned} 1. \quad I_m &= \int_0^\pi \cos(x) \cos^{m-1}(x) dx = [\sin(x) \cos^{m-1}(x)]_0^\pi + (m-1) \int_0^\pi \sin^2(x) \cos^{m-2}(x) dx \\ &\quad + \int_0^\pi \cos^{m-1}(x) \cos(x) dx \\ &\quad - (m-1) \int_0^\pi \sin(x) \cos^{m-2}(x) dx \\ &= [\sin(x) \cos^{m-1}(x)]_0^\pi + (m-1) \int_0^\pi (1 - \cos^2(x)) \cos^{m-2}(x) dx \\ &\quad + [\sin(x) \cos^{m-1}(x)]_0^\pi + (m-1) \int_0^\pi \cos^{m-2} x dx - (m-1) \int_0^\pi \cos^m(x) dx \\ &= (0 - 0) + (m-1) I_{m-2} - (m-1) I_m \end{aligned}$$

$$\therefore I_m = (m-1) I_{m-2} - (m-1) I_m$$

$$\therefore m I_m = (m-1) I_{m-2}$$

$$\therefore I_m = \frac{m-1}{m} I_{m-2}$$

$$\therefore I_4 = \frac{4-1}{4} I_2 = \frac{3}{4} \int_0^\pi \cos^2 x dx = \frac{3}{4} \int_0^\pi \frac{1 + \cos 2x}{2} dx$$

$$= \frac{3}{8} \int_0^\pi (1 + \cos 2x) dx = \frac{3}{8} \left[x + \frac{1}{2} \sin 2x \right]_0^\pi = \frac{3}{8} (\pi - 0) = \frac{3}{8} \pi$$

Question 5

Evaluate the following integrals that involve recursive relations.

1. Let

$$I_m = \int_0^\pi \cos^m(x) dx, \quad n = 0, 1, 2, \dots$$

Derive a formula relating I_m and I_{m-2} (for $m \geq 2$). Use this formula to calculate I_4 .

2. Given that

$$I_n = \int_0^\pi e^x \sin^n(x) dx, \quad n = 0, 1, 2, \dots$$

show that

$$(n^2 + 1)I_n = n(n-1)I_{n-2}.$$

Verify that

$$I_5 = \frac{3}{13}(e^\pi + 1).$$

$$\begin{aligned} 2. \quad I_n &= \int_0^\pi e^x \sin^n(x) dx = [e^x \sin^n(x)]_0^\pi - n [\cos(x) \sin^{n-1}(x) e^x]_0^\pi \\ &\quad + \int_0^\pi [-n \sin^n(x) + n(n-1) \cos^2(x) \sin^{n-2}(x)] dx \\ &\quad \begin{array}{ll} D & I \\ + \sin^n(x) & e^x \\ - n \cos(x) \sin^{n-1}(x) & e^x \\ + -n \sin^n(x) + n(n-1) \cos^2(x) \sin^{n-2}(x) & e^x \end{array} \\ &= 0 - 0 - n \int_0^\pi e^x \sin^n(x) dx \\ &\quad + n(n-1) \int_0^\pi e^x (1 - \sin^2(x)) \sin^{n-2}(x) dx \\ &= -n \int_0^\pi e^x \sin^n(x) dx + n(n-1) \int_0^\pi e^x \sin^{n-2}(x) dx \\ &\quad - n(n-1) \int_0^\pi e^x \sin^n(x) dx \\ &= -n I_n + n(n-1) I_{n-2} - n(n-1) I_n \end{aligned}$$

$$\therefore (1 + n + n^2 - n) I_n = n(n-1) I_{n-2}$$

$$\therefore (n^2 + 1) I_n = n(n-1) I_{n-2}$$

$$\therefore I_n = \frac{n(n-1)}{n^2+1} I_{n-2}$$

$$\therefore I_5 = \frac{20}{26} I_3 = \frac{10}{13} \left(\frac{6}{10} I_1 \right) = \frac{6}{13} \int_0^\pi e^x \sin(x) dx$$

$$\begin{array}{ll} D & I \\ + \sin(x) & e^x \\ - \cos(x) & e^x \\ + -\sin(x) & e^x \end{array}$$

$$\therefore \int e^x \sin(x) dx = e^x \sin(x) + e^x \cos(x) - \int e^x \sin(x) dx \quad \therefore \int e^x \sin(x) dx = \frac{1}{2} e^x \sin(x) + \frac{1}{2} e^x \cos(x)$$

$$\therefore I_5 = \frac{3}{13} [e^x \sin x + e^x \cos(x)]_0^\pi$$

$$= \frac{3}{13} (e^\pi + 1)$$

Question 6

Calculate the area delimited by the x -axis, a circle of radius $\sqrt{2}$ and the parabola $y = x^2$ in the first quadrant. See figure 1.

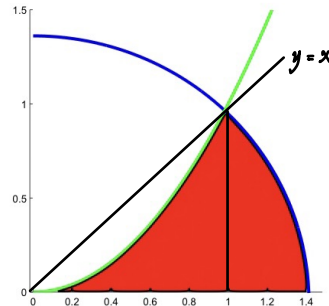


Figure 1: Area delimited by the graph of $y = x^2$, circle with radius $\sqrt{2}$ and x -axis.

$$y = x^2 = \sqrt{2-x^2} \quad \therefore x^4 = 2 - x^2 \quad \therefore x^4 + x^2 - 2 = 0 \quad \therefore (x^4 + 2x^2 - 1) = 0$$

$$\therefore x = 1 \text{ or } -1 \quad \therefore x = 1$$

$$\text{Area} = \frac{1}{8} \pi (\sqrt{2})^2 - \int_0^1 (x - x^2) dx$$

$$= \frac{1}{4} \pi - \left[\frac{1}{2} x^2 - \frac{1}{3} x^3 \right]_0^1$$

$$= \frac{1}{4} \pi - \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$= \frac{1}{4} \pi - \frac{1}{6}$$

Question 7

Calculate the volume of the solid of revolution generated by rotating the finite region bounded by the graphs of curves $y = \sqrt{x-1}$ and $y = (x-1)^2$ about the y -axis. See figure 2.

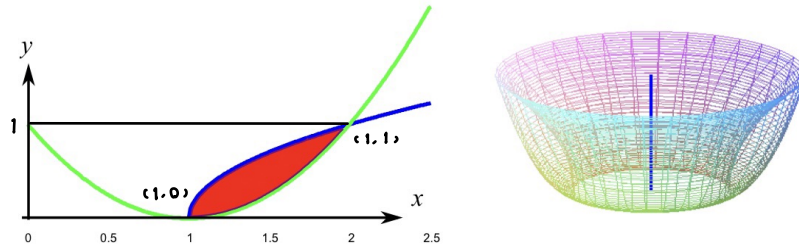


Figure 2: Area delimited by the curves $y = \sqrt{x-1}$ (blue) and $y = (x-1)^2$ (green) and the solid of revolution obtained after its rotation around the y -axis.

$$\text{If } y = \sqrt{x-1} = (x-1)^{\frac{1}{2}}$$

$$\text{Let } a = x-1 \quad \therefore a^{\frac{1}{2}} = a^{\frac{1}{2}} \quad \therefore a^{\frac{1}{2}} - a^{\frac{1}{2}} = 0 \quad \therefore a^{\frac{1}{2}}(a^{\frac{1}{2}} - 1) = 0 \quad \therefore a^{\frac{1}{2}} = 0 \text{ and } a^{\frac{1}{2}} = 1$$

$$\therefore x-1 = 0 \text{ or } 1 \quad \therefore x = 1 \text{ or } 2 \quad \therefore y = 0 \text{ or } 1$$

$$\therefore \text{intercept points } (1, 0) \text{ and } (2, 1)$$

$$y = \sqrt{x-1} = (x-1)^{\frac{1}{2}} \quad \therefore x-1 = y^2 \quad \therefore x = 1+y^2$$

$$y = (x-1)^2 \quad \therefore y^{\frac{1}{2}} = x-1 \quad \therefore x = 1+y^{\frac{1}{2}}$$

$$\therefore V = \pi \int_0^1 [(1+y^{\frac{1}{2}})^2 - (1+y^2)^2] dy$$

$$= \pi \int_0^1 (1+2y^{\frac{1}{2}}+y - 1-2y^2-y^4) dy$$

$$= \pi \int_0^1 (-y^4 - 2y^2 + y + 2y^{\frac{1}{2}}) dy$$

$$= \pi \left[-\frac{1}{5}y^5 - \frac{2}{3}y^3 + \frac{1}{2}y^2 + \frac{4}{3}y^{\frac{3}{2}} \right]_0^1$$

$$= \pi \left(\frac{29}{30} - 0 \right) = \frac{29}{30} \pi$$

Question 8

(i) Express the following complex numbers in the form $re^{i\theta}$:

$$(a) 5 - 12i \quad (b) -1 + 4i \quad (c) 2 - i \ln 3$$

(ii) Express the following complex numbers in the form $a + ib$:

$$(a) \sqrt{3}e^{\frac{i\pi}{6}} \quad (b) e^{\frac{7i\pi}{4}} \quad (c) e^{-\frac{i\pi}{4}} \quad (d) e^i$$

(i) (a) $13e^{i5.107}$

(b) $\sqrt{17}e^{i1.816}$

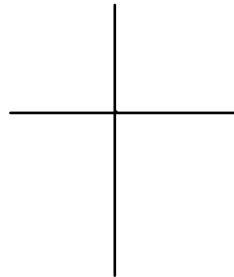
(c) $2.282e^{i5.781}$

(ii) (a) $\frac{3}{4} + \frac{\sqrt{3}}{4}i$

(b) $\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$

(c) $\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$

(d) 1



Question 9

Prove that for any complex numbers $z, w \in \mathbb{C}$,

$$(i) |z|^2 = z\bar{z}$$

$$(ii) |\operatorname{Re}(z)| \leq |z|$$

$$(iii) |z + w| \leq |z| + |w|$$

What is the geometrical interpretation of the inequality (iii)?

[Hint: for (iii) first show that $|z + w|^2 = |z|^2 + |w|^2 + \bar{w}z + \bar{z}w$]

$$z = a + ib, \bar{z} = a - ib, w = x + iy, \bar{w} = x - iy$$

$$(i) |z|^2 = (\sqrt{a^2 + b^2})^2 = a^2 + b^2$$

$$z\bar{z} = (a + ib)(a - ib) = a^2 - aib + aib + b^2 = a^2 + b^2$$

$$\therefore |z|^2 = z\bar{z}$$

$$(ii) |\operatorname{Re}(z)| = |a| = \sqrt{a^2}$$

$$|z| = \sqrt{a^2 + b^2} \quad (b^2 \geq 0)$$

$$\therefore |\operatorname{Re}(z)| - |z| \leq 0$$

$$(iii) z + w = (a + x) + i(b + y)$$

$$|z + w|^2 = (\sqrt{a^2 + x^2 + b^2 + y^2 + 2ax + 2by})^2 = a^2 + b^2 + x^2 + y^2 + 2(ax + by) \leq |z|^2 + |w|^2 + 2\operatorname{Re}(\bar{z}w)$$

$$(|z| + |w|)^2 = |z|^2 + |w|^2 + 2|z||w| = a^2 + x^2 + b^2 + y^2 + 2\sqrt{(a^2 + b^2)(x^2 + y^2)}$$

$$(|z|^2 + |w|^2 + 2|z||w|)$$

$$\therefore (|z| + |w|)^2 - |z + w|^2 = 2\sqrt{(a^2 + b^2)(x^2 + y^2)} - 2(ax + by)$$

$$\therefore (\sqrt{(a^2 + b^2)(x^2 + y^2)})^2 - (ax + by)^2 = (bx + ay)^2 \geq 0$$

$$\therefore |z + w|^2 \leq (|z| + |w|)^2$$

$$\therefore |z + w| \leq |z| + |w|$$

Question 10

Find all complex solutions to the following equation

$$1 + z + z^2 + z^3 + z^4 + z^5 = 0$$

$$\therefore z \neq 0$$

$$\therefore z + z^2 + z^3 + z^4 + z^5 + z^6 = 0$$

$$\therefore (z + z^2 + z^3 + z^4 + z^5 + z^6) - (1 + z + z^2 + z^3 + z^4 + z^5) = 0$$

$$\therefore z^6 - 1 = 0$$

$$\therefore z^6 = 1$$

$$\therefore z^6 = r(\cos \theta + i \sin \theta)$$

$$= r^6(\cos 6\theta + i \sin 6\theta) \quad \therefore r = 1$$

$$\therefore z_n = \cos \frac{2n-1\pi}{6} + i \sin \frac{2n-1\pi}{6} \quad (n \text{ from } 1 \text{ to } 6)$$

$$\therefore z_1 = \cos 0 + i \sin 0 = 1$$

$$z_2 = \cos \frac{2\pi}{6} + i \sin \frac{2\pi}{6} = \frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$z_3 = \cos \frac{4\pi}{6} + i \sin \frac{4\pi}{6} = -\frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$z_4 = \cos \frac{6\pi}{6} + i \sin \frac{6\pi}{6} = -1$$

$$z_5 = \cos \frac{8\pi}{6} + i \sin \frac{8\pi}{6} = -\frac{1}{2} - i \frac{\sqrt{3}}{2}$$

$$z_6 = \cos \frac{10\pi}{6} + i \sin \frac{10\pi}{6} = \frac{1}{2} - i \frac{\sqrt{3}}{2}$$

Question 11

Show that z is a solution to the equation

$$3 \cosh z + \sinh z = 2$$

if and only if $e^z = \frac{1}{2} \pm \frac{i}{2}$. Hence find all complex solutions to the above equation.

$$3 \left(\frac{e^z + e^{-z}}{2} \right) + \left(\frac{e^z - e^{-z}}{2} \right) = 2$$

$$\therefore 2e^z + e^{-z} = 2$$

$$\therefore 2e^z + \frac{1}{e^z} = 2$$

$$\therefore 2(e^z)^2 - 2e^z + 1 = 0$$

$$\therefore e^z = \frac{1}{2} \pm \frac{i}{2}$$

$$\therefore z = \ln \left(\frac{1}{2} \pm \frac{i}{2} \right)$$

Question 12

(i) Let $z = 2 + i$. Calculate $\arg(z)$ and $\arg(z - 1)$.

(ii) Describe the set

$$\left\{ z \in \mathbb{C} : \arg(z - 1) = \frac{\pi}{4} \right\}.$$

(iii) Describe the set

$$\left\{ z \in \mathbb{C} : \arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2} \right\}$$

$$(i) \arg(z) = \arctan\left(\frac{1}{2}\right) + 2n\pi = 0.464 + 2n\pi, \quad n = 0, \pm 1, \pm 2, \pm 3 \dots$$

$$\arg(z-1) = \arctan\left(\frac{1}{1}\right) + 2n\pi = \frac{\pi}{4} + 2n\pi, \quad n = 0, \pm 1, \pm 2, \pm 3 \dots$$

$$(ii) \arg(z-1) = \frac{\pi}{4}$$

This set of points is a half-line in the complex plane starting at -1 and extending to infinity in the direction making the angle $\theta = \frac{\pi}{4}$ with the positive real axis

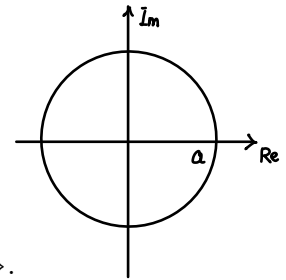
$$(iii) \arg\left(\frac{z-1}{z+1}\right) = \arg(z-1) - \arg(z+1) = \frac{\pi}{2}$$

This set of points is the difference of the angle between two half-line in the complex plane start at -1 and 1 and extending to infinity with the positive real axis

Question 13

The Joukowski transformation in aerofoil theory is

$$J(z) = z + \frac{1}{z}, \quad z \neq 0.$$



(i) Let C_1 be the circle

$$C_1 = \left\{ x + iy \in \mathbb{C} : x = a \cos \theta, y = a \sin \theta, \theta \in [0, 2\pi) \right\}.$$

of radius $a > 0$ centre the origin. Find the image of C_1 under the transformation J (i.e. the set of points $J(z)$ such that $z \in C_1$).

(ii) Find the image of the circle

$$C_2 = \left\{ x + iy \in \mathbb{C} : x = -0.05 + 1.05 \cos \theta, y = 0.05 + 1.05 \sin \theta, \theta \in [0, 2\pi) \right\}.$$

under the transformation J . Use Matlab (or otherwise) to plot the image $J(C_2)$.

$$\text{iv) } z = x + iy \quad \therefore \frac{1}{z} = \frac{1}{x + iy}$$

$$\therefore z + \frac{1}{z} = x + iy + \frac{1}{x + iy} = x + iy + \frac{x - iy}{x^2 + y^2}$$

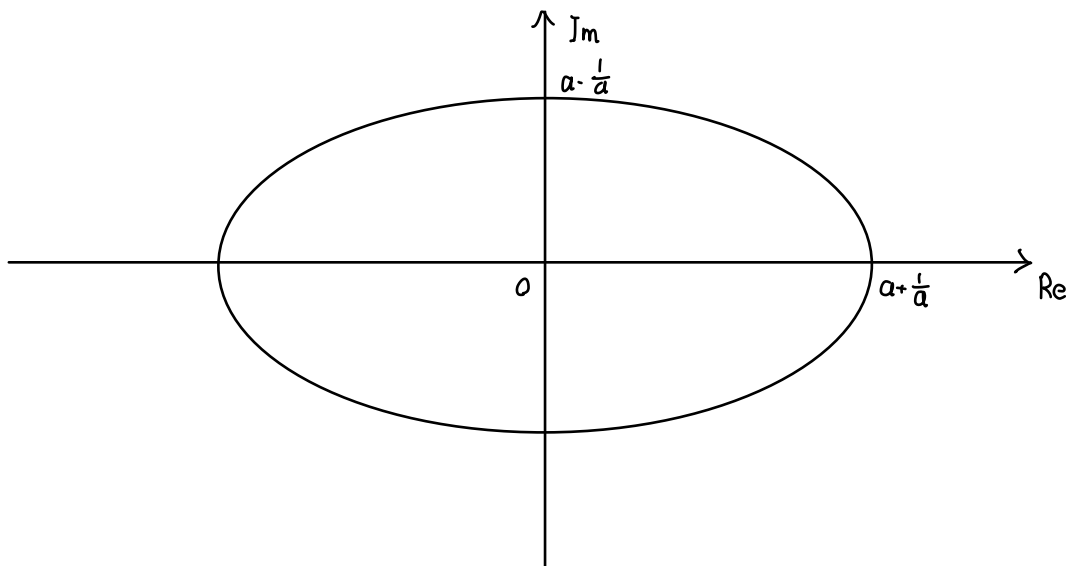
$$= \left(x + \frac{x}{x^2 + y^2} \right) + \left(y - \frac{y}{x^2 + y^2} \right) i$$

$$\therefore x = a \cos \theta, \quad y = a \sin \theta$$

$$\therefore z + \frac{1}{z} = \left(a \cos \theta + \frac{a \cos \theta}{a^2} \right) + \left(a \sin \theta - \frac{a \sin \theta}{a^2} \right) i$$

$$= \left(a + \frac{1}{a} \right) \cos \theta + i \left(a - \frac{1}{a} \right) \sin \theta \quad \therefore \frac{u^2}{\left(a + \frac{1}{a} \right)^2} + \frac{v^2}{\left(a - \frac{1}{a} \right)^2} = \cos^2 \theta + \sin^2 \theta = 1$$

$$J(C_1) = \left\{ \left(a + \frac{1}{a} \right) \cos \theta + i \left(a - \frac{1}{a} \right) \sin \theta \in \mathbb{C}, \theta \in [0, 2\pi) \right\}$$



$$\text{iv) } z = x + iy \quad \therefore \frac{1}{z} = \frac{1}{x + iy}$$

$$\therefore J(z) = z + \frac{1}{z} = x + iy + \frac{1}{x + iy}$$

$$= x + iy + \frac{x - iy}{x^2 + y^2}$$

$$= \left(x + \frac{x}{x^2 + y^2}\right) + i\left(y - \frac{y}{x^2 + y^2}\right)$$

$$\therefore x = -0.05 + 1.05 \cos \theta$$

$$y = 0.05 + 1.05 \sin \theta$$

$$\begin{aligned} \therefore x^2 + y^2 &= (0.05)^2 - 2(0.05)(1.05 \cos \theta) + (1.05 \cos \theta)^2 + (0.05)^2 + 2(0.05)(1.05 \sin \theta) + (1.05 \sin \theta)^2 \\ &= \frac{1}{200} + \frac{441}{400} + 0.105(\sin \theta - \cos \theta) = 1.1075 + 0.105(\sin \theta - \cos \theta) \end{aligned}$$

$$J(C_2) = \left\{ z + \frac{1}{z} \in \mathbb{C}, \theta \in [0, 2\pi) \right\}$$

